5D CHAOTIC MEMRISTIVE MODEL

12-Test Statistical and Performance Report

Image, Audio and Video Encryption/Decryption

Numerical case: alpha = 1 using RK4

1. Abstract

This report evaluates the five-dimensional (5D) tangent-hyperbolic memristive chaotic model in three software experiments: RGB image encryption, 16-bit PCM audio encryption, and video-frame plus audio encryption. The 12 checks cover statistical properties, chaotic dynamics, key sensitivity, key-space definition and software performance. All three experiments were rerun using the model equation. Correct-key decryption recovered every tested image pixel, audio sample, video frame and video-audio sample exactly. The encrypted data showed high measured entropy and low neighbouring-value correlation. The component-wise histogram audit found one exception: the image green channel rejects uniformity (p = 0.0086), even though pooled RGB passes. The XOR-only construction has no diffusion, so one-element NPCR/UACI checks are informational baselines, not successful avalanche tests. The effective key space is not estimated because admissible secret-key ranges and precision have not been specified.

1.1 Main findings

Question	Measured finding
Correct decryption	Exact decoded-data recovery in all three experiments; original and recovered SHA-256 hashes match.
Chaotic dynamics	Cap-like phase portraits for (x1,x2), (x1,x3) and (-x2,-x3); largest Lyapunov estimate = +1.3769.
Image and video statistics	Encrypted byte entropy is close to 8 bits and measured correlations are small. Image Test 9 passes when RGB is pooled, but the separate green-channel p-value is 0.0086 (mixed channel-wise result); sampled video frame channels pass.
Audio statistics	Encrypted sample entropy = 15.9148/16 bits; encrypted adjacent-sample correlation is about -0.002 per channel.
Plaintext-change diffusion	No diffusion: one changed input element changes only its corresponding ciphertext element under fixed XOR masks.
Effective key space	Not estimated. The current project does not define admissible parameter ranges, key precision or independent secret-key choices.
12-test status per modality	Image: 6 PASS, 0 FAIL, 5 INFO, 1 MIXED. Audio: 7 PASS, 0 FAIL, 5 INFO. Video + audio: 7 PASS, 0 FAIL, 5 INFO. MIXED is the image channel-wise Test 9 result.

Overall interpretation: the experiments demonstrate reversible multimedia processing and useful statistical scrambling in the tested samples. They do not establish cryptographic security.

2. Scope and methodology

The report follows the section order of the 12-test.

2.1 Systems and test data

Modality	Plain data actually tested	Secondary key input used
Image	test data.jpg; RGB; 250 x 334 pixels (83,500 pixels; 250,500 channel bytes).	A key image extracted from the supplied test video at t = 1.0 s and resized to 334 x 250.
Audio	test_sample-6s.wav; 44,100 Hz; stereo; 281,856 sample frames; 6.39 s; 16-bit PCM.	The audio track extracted from the supplied test video; tiled/truncated by the code to match the plain audio length.
Video + audio	test data.mp4; original 1920 x 1080, 30 fps, 12.843 s. Test preview: first 2 s, resized to 160 x 90; 60 frames; audio 48,000 Hz stereo.	A separate key-video MP4 assembled from the extracted 1.0 s video frame and the supplied WAV audio. It provides the key frame and key audio for this demonstration.

Key-media note: these secondary key assets were prepared from the test media so the optional key-image/key-audio inputs could be exercised. They are reproducible test masks, not independently generated or managed secret keys.

2.2 5D model

$$dx1/dt = a(x2 - x1) + 0.25\,x3 + g[delta + gamma\ tanh(x5)]x3$$
$$dx2/dt = b\,x1 - x1\,x3 + c\,x2 - x4$$
$$dx3/dt = d\,x1\,x2 - e\,x3$$
$$dx4/dt = x1$$
$$dx5/dt = x3$$

Parameter	Value	Parameter	Value
a	78	b	-16
c	58	d	35
e	15	g	1
delta	1	gamma	0.5
Initial state	(0, 0, 8, 6, 0)	Fractional order alpha	1

Numerical settings: classical fourth-order Runge-Kutta (RK4), step h = 0.001; first 10,000 integration steps discarded as transient; one keystream sample stored after each 10 RK4 steps. This reproduces the alpha = 1 integer-order case only. It does not implement the paper's ABC fractional numerical scheme for alpha = 0.99.

2.3 Keystream and encryption rule

The first three states (x1, x2, x3) generate the RGB byte mask. For each sampled time tn, the code forms Ki = floor(|xi(tn)| x 10^8) mod 256. For 16-bit stereo audio, x1 and x2 are converted to 16-bit masks using mod 65,536.

Image/frame: C = P XOR Kchaos XOR Kmedia. Audio: C = P XOR K16 XOR Kkey-audio. Decryption applies the same XOR masks again because XOR is its own inverse.

3. Code flow and test plan

3.1 Implementation flow

Stage	Image	Audio	Video + audio
1. Input	Read image and convert to RGB.	Read WAV as 16-bit PCM; retain sample rate and channels.	Decode video to RGB frames and extract PCM audio.
2. Chaotic mask	Run corrected 5D RK4 model; use x1, x2, x3 to build RGB byte mask.	Run same model; use x1 and x2 to build 16-bit stereo masks.	Continue one chaotic stream across frames; generate a separate stream for audio.
3. Extra key	Resize and XOR the key image.	Tile/truncate the key audio and XOR it.	Use key-video first frame as image mask and its audio track as audio mask.
4. Encrypt/decrypt	XOR original image to encrypt; repeat same masks to decrypt.	XOR PCM samples to encrypt; repeat same masks to decrypt.	XOR frames and audio separately; write/read a lossless MKV; decrypt both streams.
5. Verify	Compare every RGB pixel.	Compare every PCM sample.	Compare all processed RGB frames and PCM audio samples.

3.2 The twelve tests

No.	Test	What it checks / how to interpret it
1	Shannon entropy	How spread the encrypted values are; maximum 8 bits per byte and 16 bits per 16-bit sample.
2	Correlation coefficient	Similarity of neighbouring pixels, video frames or audio samples; near zero is preferred.
3	NPCR	Percentage of ciphertext elements changed after one plaintext element changes by one unit.
4	UACI	Average normalized ciphertext difference for the same one-element change.
5	Time-series sanity	Finite-window behaviour of the keystream states x1-x3; autocorrelation is descriptive only.
6	Phase portraits	Checks whether the first three states show the cap-like geometry in the paper.
7	Largest Lyapunov exponent	A positive numerical estimate supports chaotic sensitivity.
8	Initial-condition sensitivity	Measures changed keystream bytes after adding 1e-10 to one initial state at a time.
9	Histogram chi-square	Tests whether the encrypted byte counts are inconsistent with a uniform histogram (alpha = 0.05).
10	Key sensitivity	Tests decryption with perturbed initial conditions/parameter or incorrect media key.
11	Effective key space	Can be counted only after valid key ranges, precision and independent key choices are defined.

No.	Test	What it checks / how to interpret it
12	Performance	Measures encryption/decryption function time and throughput on this software runtime.

Important interpretation rule: the current cipher is XOR-only and has no diffusion. Therefore, for the one-element NPCR/UACI experiment, the expected NPCR is 100/N percent and UACI is at most 100/N percent. These are recorded as INFO, not compared to a 99.6%/33.5% avalanche target.

4. Results: recovery and ciphertext statistics

4.1 Decryption integrity

Modality	Processed data	Exact recovery check	Independent SHA-256 check
Image	250 x 334 RGB pixels	All pixels equal; MSE = 0.	Original/recovered pixel-array hashes identical.
Audio	281,856 stereo sample frames	All PCM samples equal; MSE = 0.	Original/recovered PCM hashes identical.
Video	60 frames at 160 x 90 plus 96,000 stereo audio sample frames	All decoded frames and audio samples equal.	Original/recovered frame and PCM audio hashes identical.

4.2 Tests 1-4: entropy, correlation, NPCR and UACI

Test	Image	Audio	Video (frames; audio)
1. Entropy	Cipher R/G/B: 7.9979 / 7.9973 / 7.9981 bits; plaintext: 7.6052 / 7.6077 / 7.0631.	Cipher 16-bit: 15.9148/16; byte entropy: 7.9999/8. Plain: 13.123/16 bits.	Frame minimum channel entropy: 7.9868/8. Video audio: 15.7314/16 bits; byte entropy 7.9995.
2. Correlation	Cipher max |r| = 0.0078; plaintext H/V/D means = 0.923/0.945/0.897.	Cipher lag-1 r = -0.0023 and -0.0024; plaintext = 0.9902 per channel.	Frame-1 spatial max |r| = 0.0100; consecutive-frame r = -0.0018; audio r = -0.0008 and 0.0001.
3. NPCR	0.0003992%; equals the one-byte XOR baseline 100/N.	0.0001774%; equals 100/N for 563,712 samples.	Frame 0.0023148%; audio 0.0005208%; each equals 100/N.
4. UACI	0.00000407%; upper bound for one changed byte = 0.0003992%.	0.00000000379%; upper bound = 0.0001774%.	Frame 0.0001059%; audio 0.0000002305%; both are within the one-element bound.

Tests 1 and 2 pass the selected checks in all three experiments. Tests 3 and 4 are INFO: the observed small values are the expected behaviour of a fixed-mask XOR construction, and show that it does not diffuse a one-element change.

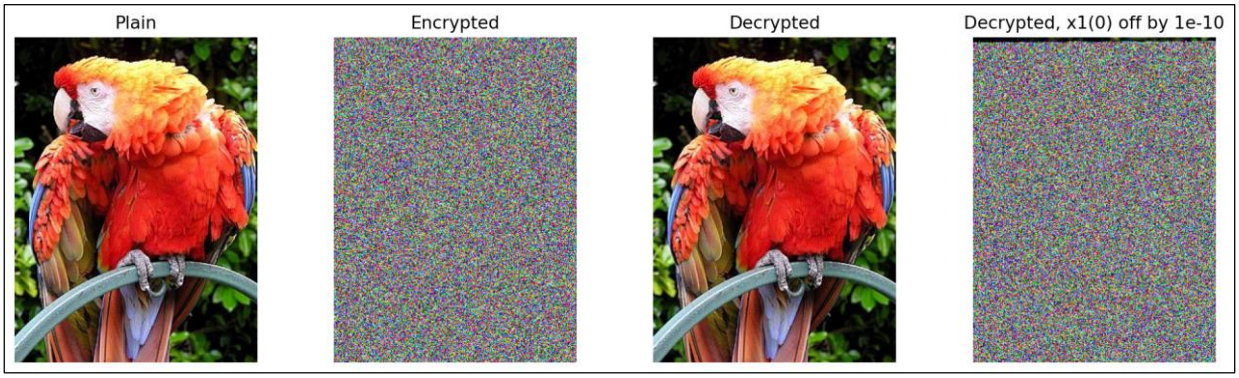


Figure 1. Supplied RGB test image: original, encrypted, correctly decrypted and decrypted with an altered initial condition.

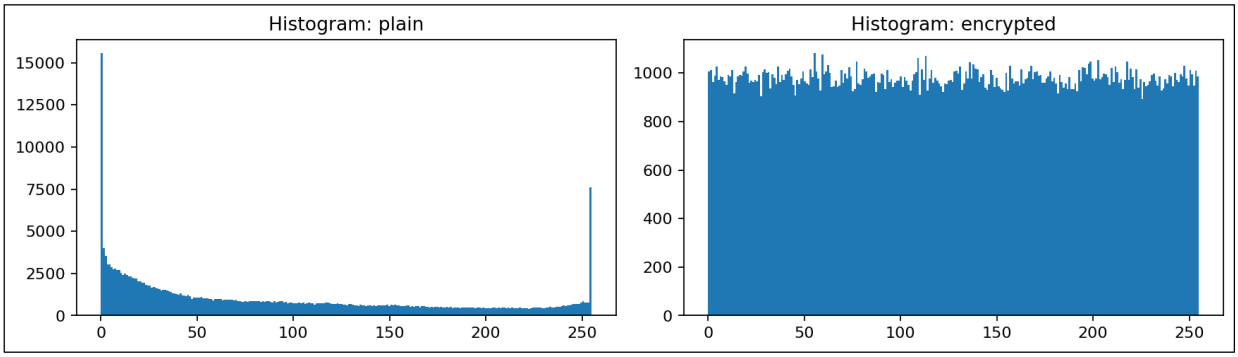


Figure 2. Original and encrypted RGB histograms. The pooled encrypted histogram looks approximately flat, but the separate green-channel chi-square test detects a deviation ($p = 0.0086$).

5. Results: chaotic generator checks (Tests 5-8)

5.1 Test 5: finite-window time-series sanity

The test code stores x1, x2 and x3 for 20,000 samples (200 time units after the transient). These keystream states remained finite, with max |x| = 127.5. The largest autocorrelation after the first zero crossing was x1 = 0.28, x2 = 0.31 and x3 = 0.39. This is a finite-window check only; it does not prove global boundedness or aperiodicity.

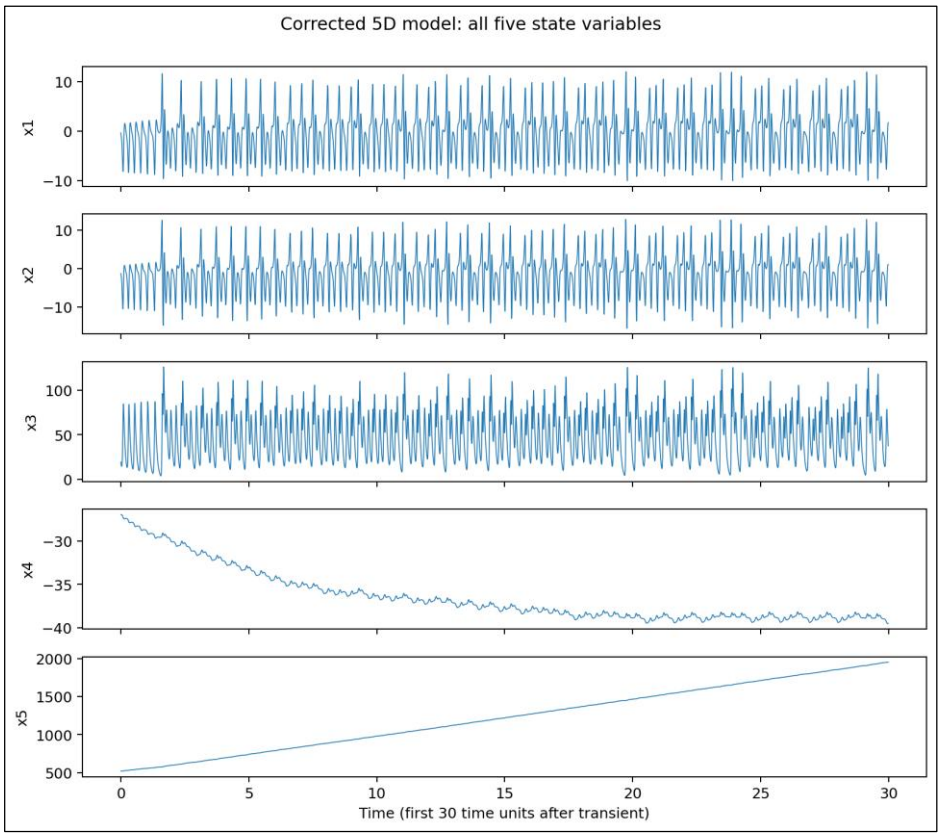


Figure 3. All five model state variables over the first 30 time units of the sampled post-transient orbit.

State	Minimum	Maximum	Note
x1	-10.046	12.062	Used in RGB/audio keystream.
x2	-15.517	12.917	Used in RGB/audio keystream.
x3	3.791	127.507	Used in RGB keystream; also plotted in the paper.
x4	-41.023	-26.969	Internal model state; not directly converted to bytes.
x5	518.573	10,256.097	Cumulative state because dx5/dt = x3; it increases over this finite window.

Important nuance: the notebook's Test 5 PASS status checks finiteness/autocorrelation of x1-x3, the three keystream states. The supplementary five-state plot shows x5 increasing throughout this window. Therefore this test must not be described as proof that every state of the full 5D system is globally bounded.

5.2 Tests 6 and 7: phase portraits and Lyapunov exponent

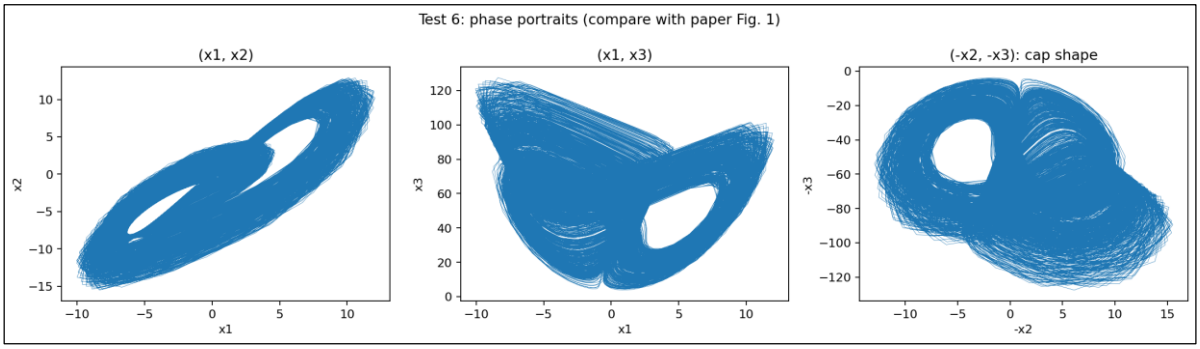


Figure 4. Phase portraits of the corrected 5D system. The (-x2,-x3) projection shows the cap-like structure.

The sampled ranges were x1 = [-10.0, 12.1], x2 = [-15.5, 12.9] and x3 = [3.8, 127.5], close to the approximate extents shown in Figure 1 of the article. The estimated largest Lyapunov exponent was +1.3769, obtained with a two-trajectory renormalization method over T = 200 time units. The article reports 0.99150 for alpha = 1. The positive sign supports chaos, but the numerical magnitude is not an exact reproduction of the article's value; it depends on the finite-time estimator and numerical setup.

5.3 Test 8: initial-condition sensitivity

Initial value perturbed by +1e-10	Image keystream bytes changed	Audio keystream bytes changed	Video keystream bytes changed
x1(0)	98.7%	99.3%	94.0%
x2(0)	98.8%	99.4%	94.7%
x3(0)	98.4%	99.3%	92.5%
x4(0)	99.4%	99.5%	98.4%
x5(0)	98.5%	99.3%	93.0%

Test 8 is marked INFO, not PASS/FAIL, because the paper does not define a numeric percentage threshold. The lower video percentages arise from its shorter 14,400-pixel-frame keystream sample. This is sensitivity to initial conditions, not evidence of a separately defined external secret key.

6. Results: histogram, key sensitivity, key space and speed (Tests 9-12)

6.1 Test 9: histogram chi-square

Modality / component	Chi-square p-value	Decision at alpha = 0.05	Interpretation
Image - Red channel	0.6737	PASS	Does not reject uniformity for the red channel.
Image - Green channel	0.0086	FAIL	Rejects uniformity for this channel. This is why the image Test 9 result is MIXED.
Image - Blue channel	0.9381	PASS	Does not reject uniformity for the blue channel.
Image - pooled RGB	0.1543	PASS (pooled only)	The pooled result does not override the green-channel finding.
Audio - low byte	0.9894	PASS	Low byte of 16-bit PCM samples; both stereo channels combined.
Audio - high byte	0.5756	PASS	High byte of 16-bit PCM samples; both stereo channels combined.
Audio - all bytes pooled	0.9960	PASS	Pooled low/high bytes and stereo channels.
Video - frame 1 R/G/B	0.3710 / 0.4402 / 0.9760	PASS (3/3)	All three channels pass on selected frame 1.
Video - frame 31 R/G/B	0.9023 / 0.8726 / 0.4408	PASS (3/3)	All three channels pass on selected middle frame.
Video - frame 60 R/G/B	0.6389 / 0.8691 / 0.6274	PASS (3/3)	All three channels pass on selected last frame.
Video audio - low/high bytes	0.8727 / 0.5311	PASS (2/2)	Both byte components pass for the extracted 48 kHz stereo audio segment.
Video + audio - all bytes pooled	0.8966	PASS (pooled)	Original combined pooled diagnostic used by the video test.

Test 9 is interpreted at alpha = 0.05. For image encryption, the pooled RGB histogram gives p = 0.1543, but the separate green-channel test gives p = 0.0086; therefore the image result is MIXED, not a clean PASS. The red and blue channels pass. Audio low/high bytes and the selected video RGB channels pass. A p-value above 0.05 means this chi-square test did not reject uniform counts; it does not prove randomness or cryptographic security. A displayed plaintext p-value of 0 is numerical underflow, not a literal probability of zero.

6.2 Test 10: wrong-key / key-sensitivity checks

Modality	Perturbation tested	Measured result
Image	x1(0) + 1e-10	NPCR 98.66%; UACI 36.69%; PSNR 6.93 dB.
Image	parameter a + 1e-10	NPCR 99.26%; UACI 36.93%; PSNR 6.90 dB.
Image	wrong key image (inverted)	NPCR 100.00%; UACI 62.82%; PSNR 3.09 dB.

Modality	Perturbation tested	Measured result
Audio	x1(0) + 1e-10	NPCR 99.67%; UACI 25.01%; SNR -16.75 dB.
Audio	parameter a + 1e-10	NPCR 99.88%; UACI 25.05%; SNR -16.75 dB.
Audio	wrong key audio (reversed)	NPCR 99.99%; UACI 8.70%; SNR -8.78 dB.
Video	x1(0) + 1e-10	Frame-1 NPCR 94.00%; UACI 35.53%; PSNR 6.97 dB; audio SNR -12.23 dB.
Video	parameter a + 1e-10	Frame-1 NPCR 97.76%; UACI 36.39%; PSNR 6.87 dB; audio SNR -12.27 dB.
Video	wrong key video (wrong frame mask/reversed audio mask)	Frame-1 NPCR 100.00%; UACI 65.53%; PSNR 2.87 dB; audio SNR -3.69 dB.

The outputs are visibly distorted under the tested wrong keys and meet the notebook's diagnostic thresholds (image/video PSNR below 15 dB and NPCR at least 90%; audio SNR below 0 dB). This demonstrates the tested key dependence, but is not a full cryptanalysis.

6.3 Test 11: effective key space

INFO - not estimated. The parameters and starting values are hard-coded in this notebook, but the project does not define permitted ranges, precision, valid chaotic regions or independent secret-key choices. The earlier 2^619 estimate is withdrawn: parameter sensitivity alone does not prove that many valid, independent, distinguishable keys exist.

6.4 Test 12: performance and throughput

Modality	Encrypt time / throughput	Decrypt time / throughput	Comment
Image	0.139 s / 1.81 MB/s	0.132 s / 1.90 MB/s	250 x 334 RGB; 0.25 MB.
Audio	0.447 s / 2.52 MB/s; 14.3x real time	0.444 s / 2.54 MB/s	6.39 s; 1.13 MB; 44.1 kHz stereo.
Video frames	1.37 s / 43.7 fps; 1.89 MB/s	1.36 s / 44.0 fps	60 frames at 160 x 90; 30 fps source; about 1.46x real-time factor.
Video audio	0.16 s; 12.8x real time	0.16 s	2.00 s; 48 kHz stereo; 0.38 MB.

Times measure the encryption/decryption functions in this software run and are environment-dependent. Video figures exclude media decoding, lossless file writing/reading and MP4 preview encoding. These are not FPGA timings.

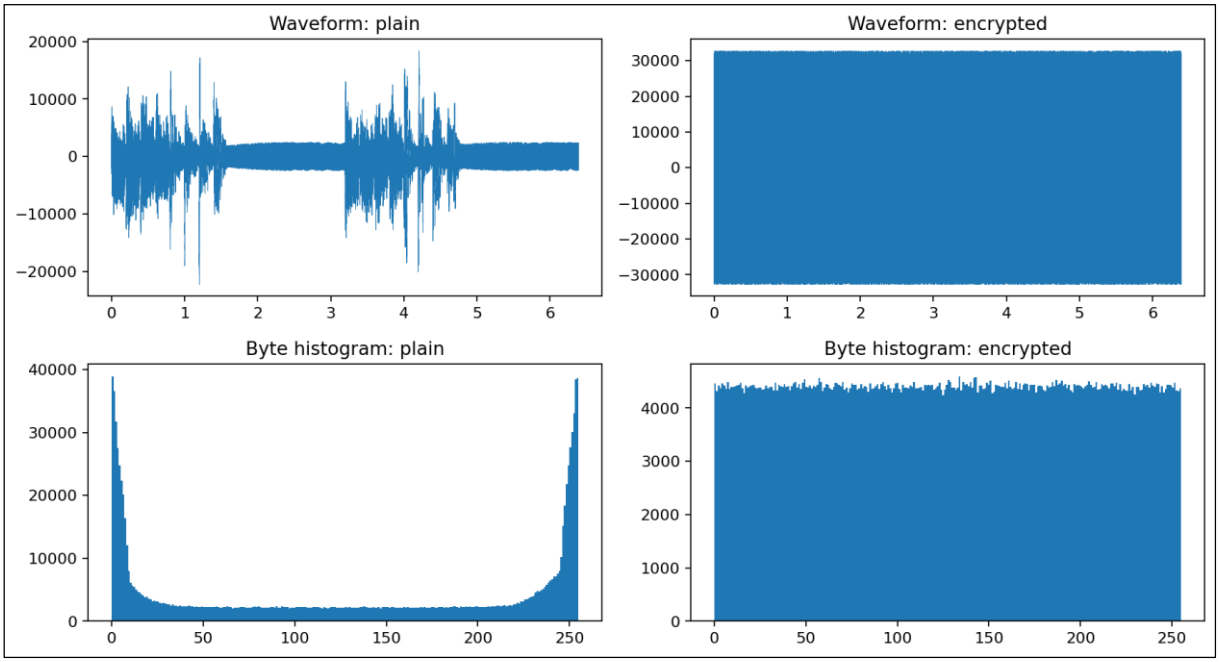


Figure 5. Audio waveform and byte-histogram outputs from the executed test.

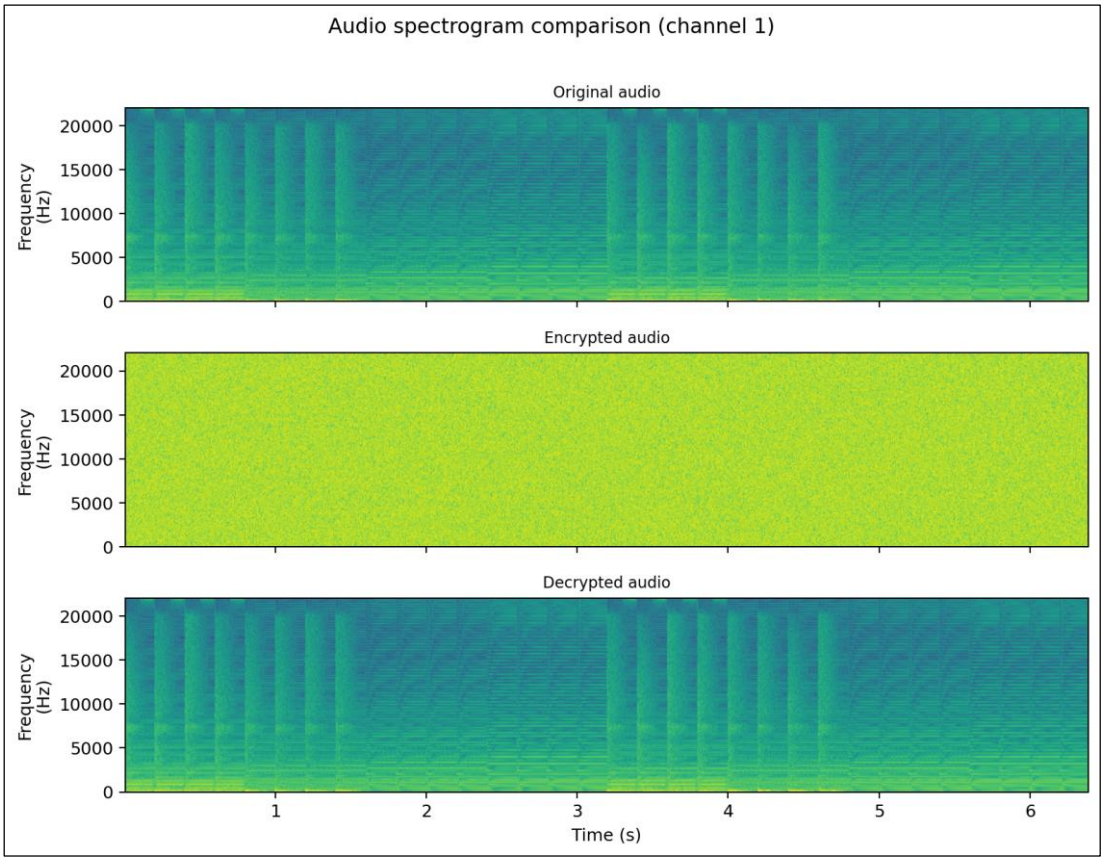


Figure 6. Original, encrypted and decrypted audio spectrograms (first channel).

7. Video outputs and cross-modality comparison

The original video was 1920 x 1080 at 30 fps and 12.843 s. To keep the 5D stream calculation practical in this run, the 12-test video experiment processed a preview segment only: 2.00 s, 160 x 90, 60 frames. The corresponding 48 kHz stereo audio segment contained 96,000 sample frames. The lossless MKV path was used for the exact recovery check.

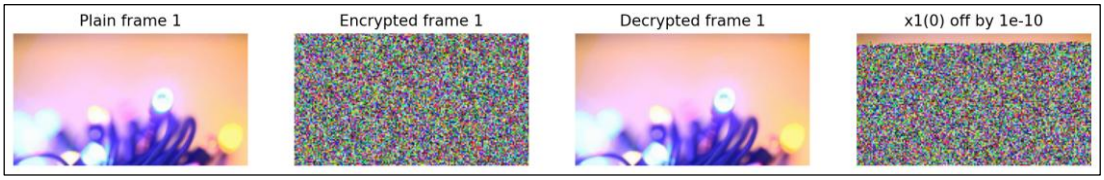


Figure 7. Video frame 1: original, encrypted, decrypted and wrong-initial-condition output.

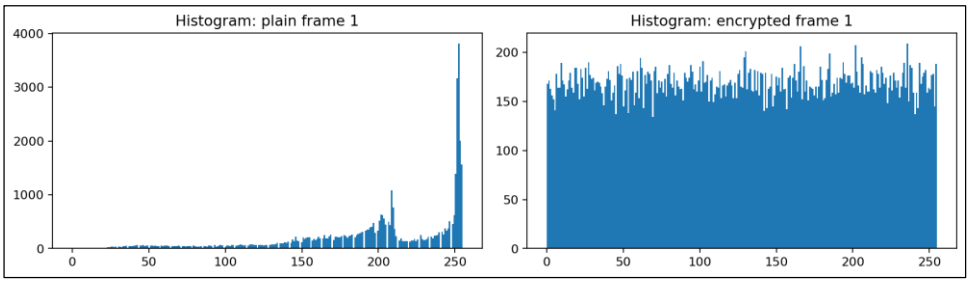


Figure 8. Original and encrypted histograms for video frame 1.

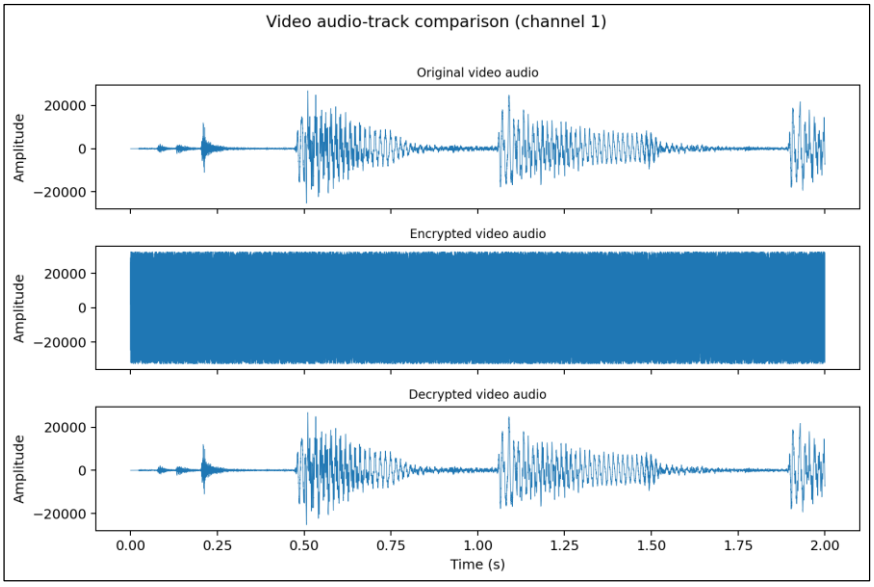


Figure 9. Video audio-track waveform comparison (original, encrypted and recovered PCM).

7.1 Cross-task comparison

Area	Image	Audio	Video + audio
Exact recovery	PASS	PASS	PASS for the 2 s / 160 x 90 preview and extracted audio.
Entropy / correlation	Strong in this sample	Strong for this WAV/key pairing	Strong for sampled frames and audio segment.
Histogram (Test 9)	MIXED: green p=0.0086; pooled RGB p=0.1543	PASS: low/high-byte p=0.9894/0.5756	PASS: 9 sampled RGB channels and audio byte components; pooled p=0.8966
NPCR / UACI	No diffusion; INFO baseline	No diffusion; INFO baseline	No diffusion; INFO baseline.
Generator tests	Same 5D orbit diagnostics	Same 5D orbit diagnostics	Same 5D orbit diagnostics; Test 8 sample is shorter.
Key space	Not estimated	Not estimated	Not estimated.
Benchmark	1.81 MB/s encryption	14.3x real-time audio encryption	43.7 fps preview-frame encryption; 1.46x source frame rate.

8. Discussion and conclusion

- Exact recovery is verified. The recovered image pixels, audio PCM samples, video RGB frames and video PCM samples match the processed originals. SHA-256 checks of decoded data also match.
- Image/video byte statistics improve strongly in this sample. Entropy is close to the maximum and measured neighbouring-value correlations are small. The image histogram is not uniform in every channel: green has p=0.0086, although pooled RGB, red and blue pass. Audio low/high-byte checks and the sampled video channel checks pass.
- The construction has no diffusion. A one-element plaintext change changes only the corresponding ciphertext element under the same fixed masks. This is why Tests 3 and 4 are INFO and why the system should not be described as having an avalanche effect.
- The key space remains unknown. A positive Lyapunov exponent and high sensitivity do not define an external cryptographic key or a key-space size. Do not report 2^619.
- The video result is a preview experiment. It does not prove full-resolution, full-duration performance or recovery of the original compressed MP4 bitstream. Exactness applies to the processed RGB frames and extracted PCM samples in the lossless path.

8.1 Final conclusion

The 5D model provides a reproducible chaotic source for the tested image, audio and video XOR pipelines. All correct-key round trips were exact, and the tested ciphertext statistics were favourable. The current design is best presented as an experimental multimedia encryption prototype, not as a fully secure cipher: it lacks diffusion, has no defensible key-space estimate and uses a deterministic keystream. A future secure design would require a defined external secret key, key-stream non-reuse, a diffusion/authenticated-encryption stage, and broader cryptanalysis.

8.2 Scope limitations

- The paper's direct encryption demonstration is the amalgamated image algorithm. Audio and video are extensions in this project.
- The report uses RK4 at alpha = 1; it does not reproduce the paper's ABC fractional-order alpha = 0.99 calculation.
- The Lyapunov estimate +1.3769 is positive but differs from the article's 0.99150; it is labelled as this implementation's finite-time estimate.
- The video 12-test run processes only 2 seconds at 160 x 90; do not claim full input-video testing.

8.3 Files included in the companion package

- Corrected 12-test notebook and CSV result tables for image, audio and video.
- Original test inputs and generated key-media files used in this rerun.
- Original, encrypted, decrypted and wrong-key outputs; lossless encrypted/decrypted video containers; viewable MP4 previews.
- Time-series, phase-portrait, image histogram, audio waveform/spectrogram, video frame/histogram and video-audio waveform figures.
- Independent SHA-256 and decoded-data equality verification summary.

References

Qureshi, M. A., Khan, N. A., Raza, S. and Iqbal, S. M. Z. (2024). "Cap like trajectories in 5D chaotic tangent hyperbolic memristive model: fractional calculus and encryption." Physica Scripta, 99, 075238. https://doi.org/10.1088/1402-4896/ad5476.

Github link:

https://github.com/fizza-14/5D-Chaotic-Memristive-Image-Audio-Video-Encryption-Decryption